

T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content $\text{SM}+S_1+S_2$. The report combines the renormalisable RGE running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F . Throughout, $16\pi^2\mu dX/d\mu=\beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Renormalisable EFT1 couplings

$\beta_{g_Y}^{(1)}$

$$16\pi^2\mu\frac{dg_Y}{d\mu}=\beta_{g_Y}^{(1)}$$

Column Representative term structure	
T1	$\frac{47g_Y^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{47g_Y^3}{6}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{15g_Y^3}{2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{15g_Y^3}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{43g_Y^3}{6}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{47g_Y^3}{6}$

$$\beta_{g_2}^{(1)}$$

$$16\pi^2\,\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{5g_2^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5g_2^3}{2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{17g_2^3}{6}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{17g_2^3}{6}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{5g_2^3}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11g_2^3}{6}$

$$\beta_{g_3}^{(1)}$$

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column Representative term structure

T1	$-7g_3^3$
----	-----------

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-7g_3^3$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-7g_3^3$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$

$\beta_{Y_u}^{(1)}$

$16\pi^2\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$
T2	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$
T3	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$
T3	$-8g_3^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Column	Representative term structure
T1	$-\frac{17}{12}gY^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

$16\pi^2\,\mu\,\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}gY^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

$16\pi^2\,\mu\,\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 2

Column	Representative term structure	
T1	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	
T2	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	
T3	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$
T3	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$

$$\beta_{m_{S_1}^2}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dm_{S_1}^2}{d\mu}=\beta_{m_{S_1}^2}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$6\lambda_{12}m_{S_2}^2$
T2	$+4\lambda_{S_1}m_{S_1}^2$
T3	$-\frac{9g_2^2m_{S_1}^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$6\lambda_{12}m_{S_2}^2$	$+4\lambda_{S_1}m_{S_1}^2$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}m_{S_2}^2$	$+6\lambda_{S_1}m_{S_1}^2$	$-\frac{9g_2^2m_{S_1}^2}{2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}m_{S_2}^2$	$+6\lambda_{S_1}m_{S_1}^2$	$-\frac{9g_2^2m_{S_1}^2}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}m_{S_2}^2$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$-12g_2^2m_{S_1}^2$
T2	$+m_{S_1}^2(8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$
T3	$+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12g_2^2m_{S_1}^2$	$+m_{S_1}^2(8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12g_2^2m_{S_1}^2$	$+m_{S_1}^2(8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$	$+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{m_{S_2}^2}^{(1)}$

$16\pi^2\mu\frac{dm_{S_2}^2}{d\mu}=\beta_{m_{S_2}^2}^{(1)}$

Terms 1 of 3

Column Representative term structure	
T1	$-12g_2^2m_{S_2}^2$
T2	$-6g_Y^2m_{S_2}^2$
T3	$+2\lambda_{12}m_{S_1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12g_2^2m_{S_2}^2$	$-6g_Y^2m_{S_2}^2$	$+2\lambda_{12}m_{S_1}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-6g_Y^2m_{S_2}^2$	$+4\lambda_{12}m_{S_1}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-6g_Y^2m_{S_2}^2$	$+4\lambda_{12}m_{S_1}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-6g_Y^2m_{S_2}^2$	$+6\lambda_{12}m_{S_1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12g_2^2m_{S_2}^2$	$-6g_Y^2m_{S_2}^2$	—

Terms 2 of 3

Column	Representative term structure
T1	$+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$
T2	$-\frac{9g_2^2m_{S_2}^2}{2}$
T3	$+6\lambda_{S_2}m_{S_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{9g_2^2m_{S_2}^2}{2}$	$+6\lambda_{S_2}m_{S_2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{9g_2^2m_{S_2}^2}{2}$	$+6\lambda_{S_2}m_{S_2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$+4\lambda_{S_2}m_{S_2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	—	—

Terms 3 of 3

Column	Representative term structure
T1	$+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_H}^{(1)}$

$16\pi^2\mu\frac{d\lambda_H}{d\mu}=\beta_{\lambda_H}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$
T3	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$

Terms 2 of 6

Column	Representative term structure
T1	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$
T2	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$

Terms 3 of 6

Column	Representative term structure
T1	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$
T2	$-9g_2^2\lambda_H$
T3	$+\frac{3g_Y^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$

Terms 4 of 6

Column	Representative term structure
T1	$+12\lambda_H^2$
T2	$+2\lambda_{H1}^2$
T3	$+6\lambda_{H2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+2\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+2\lambda_{H2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+6\lambda_{H2}^2$

Terms 5 of 6

Column	Representative term structure
T1	$+3\lambda_{H2}^{\text{Adj}^2}$
T2	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$
T3	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—

Terms 6 of 6

Column	Representative term structure
T1	$+3\lambda_{H1}^{\text{Adj}^2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$

$\beta_{\lambda_{S_1}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}}{d\mu}=\beta_{\lambda_{S_1}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$6\lambda_{12}^2$
T2	$+4\lambda_{H1}^2$
T3	$+10\lambda_{S_1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$6\lambda_{12}^2$	$+4\lambda_{H1}^2$	$+10\lambda_{S_1}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$	$+12\lambda_{S_1}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$	$+12\lambda_{S_1}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}^2$	$+4\lambda_{H1}^2$	$+14\lambda_{S_1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$	$+14\lambda_{S_1}^2$

Terms 2 of 5

Column	Representative term structure
T1	$\frac{9g_2^4}{4}$
T2	$-9g_2^2\lambda_{S_1}$
T3	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$
T2	$21g_2^4$
T3	$+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$21g_2^4$	$+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$21g_2^4$	$+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$

Terms 4 of 5

Column	Representative term structure	
T1	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$	
T2	$+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$	
T3	$+26\lambda_{S1}^{\text{Adj}^2}$	

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$	$+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$	$+26\lambda_{S1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$	$+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$	$+26\lambda_{S1}^{\text{Adj}^2}$

Terms 5 of 5

Column	Representative term structure
T1	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T2	$+6\lambda_{12}^{\text{Adj}^2}$
T3	$+6\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$

$\beta_{\lambda_{S_2}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$

Terms 1 of 7

Column	Representative term structure
T1	$21g_2^4$
T2	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$
T3	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$21g_2^4$	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$21g_2^4$	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$

Terms 2 of 7

Column	Representative term structure
T1	$+12g_Y^4$
T2	$+2\lambda_{12}^2$
T3	$+4\lambda_{H2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12g_Y^4$	$+2\lambda_{12}^2$	$+4\lambda_{H2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+12g_Y^4$	$+4\lambda_{12}^2$	$+4\lambda_{H2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+12g_Y^4$	$+4\lambda_{12}^2$	$+4\lambda_{H2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$12g_Y^4$	$+6\lambda_{12}^2$	$+4\lambda_{H2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12g_Y^4$	$+6\lambda_{12}^2$	$+4\lambda_{H2}^2$

Terms 3 of 7

Column	Representative term structure
T1	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$
T2	$+14\lambda_{S_2}^2$
T3	$+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S_2}^2$	$+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+12\lambda_{S_2}^2$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+12\lambda_{S_2}^2$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+10\lambda_{S_2}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S_2}^2$	$+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$

Terms 4 of 7

Column	Representative term structure
T1	$+26\lambda_{S_2}^{\text{Adj}^2}$
T2	$\frac{9g_2^4}{4}$
T3	$+g_Y^2\left(6g_2^2-12\lambda_{S_2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$\frac{9g_2^4}{4}$	$+g_Y^2\left(6g_2^2-12\lambda_{S_2}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{9g_2^4}{4}$	$+g_Y^2\left(6g_2^2-12\lambda_{S_2}\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	—	—

Terms 5 of 7

Column	Representative term structure
T1	$-9g_2^2\lambda_{S_2}$
T2	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 7

Column	Representative term structure
T1	$-12g_Y^2\lambda_{S_2}$
T2	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3	$+6\lambda_{12}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12g_Y^2\lambda_{S_2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+6\lambda_{12}^{\text{Adj}^2}$

Terms 7 of 7

Column	Representative term structure
T1	$+6\lambda_{12}^{\text{Cross}2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{\text{Cross}2}$

$\beta_{\lambda_{H1}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{H1}}{d\mu}=\beta_{\lambda_{H1}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T2	$+12\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T3	$-\frac{9g_2^2\lambda_{H1}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+12\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$-\frac{9g_2^2\lambda_{H1}}{2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+6\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+6\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$-\frac{33g_2^2\lambda_{H1}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$-\frac{33g_2^2\lambda_{H1}}{2}$

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{3g_Y^2\lambda_{H1}}{2}$
T2	$+6\lambda_{12}\lambda_{H2}$
T3	$+6\lambda_H\lambda_{H1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	$+6\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	$+2\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	—	$+6\lambda_H\lambda_{H1}$

Terms 3 of 5

Column	Representative term structure
T1	$+4\lambda_{H1}^2$
T2	$+4\lambda_{H1}\lambda_{S1}$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+4\lambda_{H1}\lambda_{S1}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+6\lambda_{H1}\lambda_{S1}$	$+\frac{9g_2^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+6\lambda_{H1}\lambda_{S1}$	$+\frac{9g_2^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+8\lambda_{H1}\lambda_{S1}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+8\lambda_{H1}\lambda_{S1}$	—

Terms 4 of 5

Column	Representative term structure
T1	$-9g_2^2\lambda_{H1}$
T2	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1}^{\text{Adj}})$

Terms 5 of 5

Column	Representative term structure
T1	$+6g_2^4$
T2	$+6\lambda_{H1}^{\text{Adj}^2}$
T3	$+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+6g_2^4$	$+6\lambda_{H1}^{\text{Adj}^2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$+6\lambda_{H1}^{\text{Adj}^2}$	$+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_{H2}}^{(1)}$

$16\pi^2\,\mu\,\frac{d\lambda_{H2}}{d\mu}=\beta_{\lambda_{H2}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{H2}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$
T2	$+4\lambda_{T3}\,(\lambda_{T3})^*$
T3	$+6g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$	$+4\lambda_{T3}\,(\lambda_{T3})^*$	$+6g_2^4$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+6\lambda_{T3}\,(\lambda_{T3})^*$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+6\lambda_{T3}\,(\lambda_{T3})^*$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+12\lambda_{T3}\,(\lambda_{T3})^*$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$	$+4\lambda_{T3}\,(\lambda_{T3})^*$	$+6g_2^4$

Terms 2 of 6

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{H2}}{2}$
T2	$+3g_Y^4$
T3	$-\frac{15g_Y^2\lambda_{H2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{9g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H2}}{2}$

Terms 3 of 6

Column	Representative term structure
T1	$+2\lambda_{12}\lambda_{H1}$
T2	$+6\lambda_H\lambda_{H2}$
T3	$+4\lambda_{H2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+6\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$

Terms 4 of 6

Column	Representative term structure
T1	$+8\lambda_{H2}\lambda_{S2}$
T2	$+6\lambda_{H2}^{\text{Adj}^2}$
T3	$\lambda_{H2}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+6\lambda_{H2}\lambda_{S2}$	—	$\lambda_{H2}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+6\lambda_{H2}\lambda_{S2}$	—	$\lambda_{H2}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{S2}$	—	$\lambda_{H2}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—

Terms 5 of 6

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$-9g_2^2\lambda_{H2}$
T3	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_{12}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 8

Column	Representative term structure	
T1	$4\lambda_{T3}\left(\lambda_{T3}\right)^*$	
T2	$-12g_2^2\lambda_{12}$	
T3	$-6g_Y^2\lambda_{12}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$4\lambda_{T3}\left(\lambda_{T3}\right)^*$	$-12g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$3\lambda_{T3}\left(\lambda_{T3}\right)^*$	$-9g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$3\lambda_{T3}\left(\lambda_{T3}\right)^*$	$-9g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$4\lambda_{T3}\left(\lambda_{T3}\right)^*$	$-12g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 8

Column	Representative term structure
T1	$+4\lambda_{12}^2$
T2	$+4\lambda_{12}\lambda_{S_1}$
T3	$+8\lambda_{12}\lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}^2$	$+4\lambda_{12}\lambda_{S_1}$	$+8\lambda_{12}\lambda_{S_2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}^2$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}^2$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{12}^2$	$+8\lambda_{12}\lambda_{S_1}$	$+4\lambda_{12}\lambda_{S_2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}^2$	—	—

Terms 3 of 8

Column	Representative term structure
T1	$+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$
T2	$+4\lambda_{H1}\lambda_{H2}$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$	$+4\lambda_{H1}\lambda_{H2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+4\lambda_{H1}\lambda_{H2}$	$+\frac{9g_2^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+4\lambda_{H1}\lambda_{H2}$	$+\frac{9g_2^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+4\lambda_{H1}\lambda_{H2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{H1}\lambda_{H2}$	—

Terms 4 of 8

Column	Representative term structure
T1	$+ \frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T2	$+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$
T3	$\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+ \frac{3\lambda_{12}^{\text{Adj}^2}}{4}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+ \frac{3\lambda_{12}^{\text{Adj}^2}}{4}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$

Terms 5 of 8

Column	Representative term structure
T1	$+16g_2^4$
T2	$+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$
T3	$+g_Y^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16g_2^4$	$+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$	$+g_Y^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$

Terms 6 of 8

Column	Representative term structure
T1	$+ \lambda_{S_1} \left(8 \lambda_{12} + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$
T2	$+ \lambda_{12} \left(\frac{16 \lambda_{12}^{\text{Cross}}}{3} + 4 \lambda_{S_1}^{\text{Adj}} + 4 \lambda_{S_2}^{\text{Adj}} \right)$
T3	$+ \lambda_{S_2} \left(8 \lambda_{12} + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \lambda_{S_1} \left(8 \lambda_{12} + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$	$+ \lambda_{12} \left(\frac{16 \lambda_{12}^{\text{Cross}}}{3} + 4 \lambda_{S_1}^{\text{Adj}} + 4 \lambda_{S_2}^{\text{Adj}} \right)$	$+ \lambda_{S_2} \left(8 \lambda_{12} + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$

Terms 7 of 8

Column	Representative term structure
T1	$+16\lambda_{12}^{\text{Adj}^2}$
T2	$+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$
T3	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16\lambda_{12}^{\text{Adj}^2}$	$+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$

Column	Representative term structure
T1	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$

$\beta_{\lambda_{T3}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{T3}}{d\mu}=\beta_{\lambda_{T3}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{T3}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{H2}^{\text{Adj}})$
T2	$-\frac{21g_2^2\lambda_{T3}}{2}$
T3	$-\frac{9g_Y^2\lambda_{T3}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{T3}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{H2}^{\text{Adj}})$	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{33g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$

Terms 2 of 4

Column	Representative term structure
T1	$+2\lambda_{12}\lambda_{T3}$
T2	$+2\lambda_H\lambda_{T3}$
T3	$+4\lambda_{H1}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$

Terms 3 of 4

Column	Representative term structure
T1	$+4\lambda_{H2}\lambda_{T3}$
T2	$\lambda_{T3}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T3	$-9g_2^2\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	$\lambda_{T3}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{T3}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	$\lambda_{T3}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{T3}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\lambda_{T3}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)-4\lambda_{H1}^{\text{Adj}})$
T2	$\lambda_{T3}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)-\frac{4\lambda_{12}^{\text{Adj}}}{3}-2\lambda_{12}^{\text{Cross}}-2\lambda_{H1}^{\text{Adj}}-\frac{4\lambda_{H2}^{\text{Adj}}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{T3}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)-4\lambda_{H1}^{\text{Adj}})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{T3}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)-\frac{4\lambda_{12}^{\text{Adj}}}{3}-2\lambda_{12}^{\text{Cross}}-2\lambda_{H1}^{\text{Adj}}-\frac{4\lambda_{H2}^{\text{Adj}}}{3}\right)$

$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T2	$-9g_2^2\lambda_{12}^{\text{Adj}}$
T3	$-6g_Y^2\lambda_{12}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2}{9}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$-24g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T2	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$
T3	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T2	$+\frac{16\lambda_{12}^{\text{Adj}}\lambda_{12}^{\text{Cross}}}{3}$
T3	$+16\lambda_{12}^{\text{Adj}}\lambda_{S1}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{16\lambda_{12}^{\text{Adj}}\lambda_{12}^{\text{Cross}}}{3}$	$+16\lambda_{12}^{\text{Adj}}\lambda_{S1}^{\text{Adj}}$

Terms 4 of 4

Column	Representative term structure
T1	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$
T2	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T2	$+\frac{33g_2^4}{4}$
T3	$+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$

Terms 2 of 5

Column	Representative term structure	
T1	$+g_Y{}^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$	
T2	$+\lambda_{12}{}^2$	
T3	$+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$	$+\lambda_{12}{}^2$	$+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$

Terms 3 of 5

Column	Representative term structure
T1	$+\lambda_{12}(8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}})$
T2	$+\lambda_{S_2}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$
T3	$+4\lambda_{12}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{12}(8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}})$	$+\lambda_{S_2}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$	$+4\lambda_{12}^{\text{Adj}^2}$

Terms 4 of 5

Column	Representative term structure
T1	$+11\lambda_{12}^{\text{Cross}^2}$
T2	$-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$
T3	$-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+11\lambda_{12}^{\text{Cross}^2}$	$-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$	$-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$

Terms 5 of 5

Column	Representative term structure
T1	$+ \lambda_{H1} \lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \lambda_{H1} \lambda_{H2}$

$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{H1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H1}^{\text{Adj}}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\right)$
T2	$+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T3	$-9g_2^2\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\right)$	$+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$-9g_2^2\lambda_{H1}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\right)$	$+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$-9g_2^2\lambda_{H1}^{\text{Adj}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	—

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$
T2	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$

Terms 3 of 4

Column	Representative term structure
T1	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$
T2	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$
T3	$\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S1}^{\text{Adj}})$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	$\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	$\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S1}^{\text{Adj}})$

Terms 4 of 4

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T2	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{H2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S2}^{\text{Adj}})$
T2	$+\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T3	$+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$+g_Y^2\left(12g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$+g_Y^2\left(12g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S2}^{\text{Adj}})$	$-\frac{8}{9}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$
T2	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$
T2	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T3	$+16\lambda_{T3}\left(\lambda_{T3}\right)^{*}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+16\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+16\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—	—

Terms 4 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{H2}^{\text{Adj}}$
T2	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3	$+12\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+12\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$

$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{3g_2^4}{4}$
T2	$+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$
T3	$+\frac{\lambda_{12}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$	$+\frac{\lambda_{12}^2}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$	—

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{\lambda_{H1}^2}{2}$
T2	$+3\lambda_{H1}^{\text{Adj}2}$
T3	$+\frac{7\lambda_{S1}^2}{4}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{\lambda_{H1}^2}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{\lambda_{H1}^2}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$

Terms 3 of 4

Column	Representative term structure
T1	$+13\lambda_{S_1}\lambda_{S_1}^{\text{Adj}}$
T2	$+22\lambda_{S_1}^{\text{Adj}^2}$
T3	$+\frac{3\lambda_{12}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+13\lambda_{S_1}\lambda_{S_1}^{\text{Adj}}$	$+22\lambda_{S_1}^{\text{Adj}^2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+13\lambda_{S_1}\lambda_{S_1}^{\text{Adj}}$	$+22\lambda_{S_1}^{\text{Adj}^2}$	$+\frac{3\lambda_{12}^2}{4}$

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$
T2	$+12\lambda_{12}^{\text{Adj}^2}$
T3	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+12\lambda_{12}^{\text{Adj}^2}$	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

$\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{3g_2^4}{4}$
T2	$+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$
T3	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3g_2^4}{4}$	$+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3g_2^4}{4}$	$+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$

Terms 2 of 5

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{2}$
T2	$+ \frac{\lambda_{12}^2}{4}$
T3	$+ \frac{\lambda_{H2}^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_Y^4}{2}$	$+ \frac{\lambda_{12}^2}{4}$	$+ \frac{\lambda_{H2}^2}{2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_Y^4}{2}$	—	$+ \frac{\lambda_{H2}^2}{2}$

Terms 3 of 5

Column	Representative term structure
T1	$+3\lambda_{H2}^{\text{Adj}^2}$
T2	$+\frac{7\lambda_{S2}^2}{4}$
T3	$+13\lambda_{S2}\lambda_{S2}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$	$+13\lambda_{S2}\lambda_{S2}^{\text{Adj}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$	$+13\lambda_{S2}\lambda_{S2}^{\text{Adj}}$

Terms 4 of 5

Column	Representative term structure
T1	$+22\lambda_{S_2}^{\text{Adj}^2}$
T2	$+\frac{3\lambda_{12}^2}{4}$
T3	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$

Terms 5 of 5

Column	Representative term structure
T1	$+12\lambda_{12}^{\text{Adj}^2}$
T2	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}^2}$	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab} . Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$$\beta_{C_{1133}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1133}}{d\mu}=\beta_{C_{1133}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1134}}^{(1)}$

$16\pi^2\mu\frac{dC_{1134}}{d\mu}=\beta_{C_{1134}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}i\lambda T_3\overline{y_1y_2}}{MF}$
T2	$\frac{i\lambda T_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}i\lambda T_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda T_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda T_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{2i\lambda T_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda T_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-\frac{2\sqrt{6}i\lambda T_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1144}}^{(1)}$

$16\pi^2\mu\frac{dC_{1144}}{d\mu}=\beta_{C_{1144}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1157}}^{(1)}$

$$16\pi^2\,\mu\,\frac{dC_{1157}}{d\mu}=\beta_{C_{1157}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\,12\,Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\,12\,Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2y\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2y\overline{e}}{12MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{1158}}^{(1)}$

$16\pi^2\mu\frac{dC_{1158}}{d\mu}=\beta_{C_{1158}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{11511}}^{(1)}$

$16\pi^2\mu\frac{dC_{11511}}{d\mu}=\beta_{C_{11511}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T3	$\frac{g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3 M F}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3 g_2^2 \overline{y_1 y_2}}{4 M F}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3 M F}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 6

Column	Representative term structure
T1	$\frac{g_1^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T3	$\frac{y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3 g_1^2 \overline{y_1 y_2}}{8 M F}$	$-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3 M F}$	$\frac{y e \overline{y_1 y_2 y e}}{4 M F}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 6

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{MF}$
T2	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{11512}}^{(1)}$

$$16\pi^2\,\mu\frac{dC_{11512}}{d\mu}=\beta_{C_{11512}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda\,Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda\,Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure
T 1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2}ye}{MF}$
T 2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T 3	$\frac{ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T 1	T 2	T 3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2}ye}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 6

Column	Representative term structure
T1	$\frac{ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3ig_1^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1}y_2}{3MF}$	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{4MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 6

Column	Representative term structure
T1	$\frac{i\lambda d a_{12} I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{3} i g_1^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{i\lambda d a_{12} I n v 2 \overline{y_1 y_2}}{3 M F}$	$\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{4 M F}$	$\frac{\sqrt{3} i g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y \bar{e}}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3} i y e \overline{y_1 y_2 y \bar{e}}}{12 M F}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$	$-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{11513}}^{(1)}$

$16\pi^2\mu\frac{dC_{11513}}{d\mu}=\beta_{C_{11513}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11\sqrt{3}g_2^2y_1y_2}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3\lambda da12Inv1\overline{y_1y_2}}}{MF}$
T2	$\frac{\sqrt{3ig_2^2\overline{y_1y_2}}}{MF}$
T3	$\frac{\sqrt{3ye\overline{y_1y_2ye}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3\lambda da12Inv1\overline{y_1y_2}}}{6MF}$	$-\frac{\sqrt{3ig_2^2\overline{y_1y_2}}}{6MF}$	$\frac{\sqrt{3ye\overline{y_1y_2ye}}}{12MF}$

$\beta_{C_{11514}}^{(1)}$

$$16\pi^2\mu\frac{dC_{11514}}{d\mu}=\beta_{C_{11514}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{1167}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1167}}{d\mu}=\beta_{C_{1167}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{1168}}^{(1)}$

$16\pi^2\mu\frac{dC_{1168}}{d\mu}=\beta_{C_{1168}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2y\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2y\overline{e}}{12MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{11611}}^{(1)}$

$16\pi^2\mu\frac{dC_{11611}}{d\mu}=\beta_{C_{11611}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure
T 1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$
T 2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T 3	$\frac{ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T 1	T 2	T 3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 6

Column	Representative term structure
T1	$\frac{ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{3ig_1^2\overline{y_1}y_2}{8MF}$	$-\frac{iy_e\overline{y_1}y_2y_e}{4MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1}y_2}{3MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 6

Column	Representative term structure
T1	$\frac{i\lambda d_{12} I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{3} i g_1^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{i\lambda d_{12} I n v 2 \overline{y_1 y_2}}{3 M F}$	$-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{4 M F}$	$-\frac{\sqrt{3} i g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y \bar{e}}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y \bar{e}}}{12 M F}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{11612}}^{(1)}$

$$16\pi^2\mu\frac{dC_{11612}}{d\mu}=\beta_{C_{11612}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T3	$\frac{g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3 M F}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3 g_2^2 \overline{y_1 y_2}}{4 M F}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3 M F}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 6

Column	Representative term structure
T1	$\frac{g_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{MF}$
T3	$\frac{y e \overline{y_1 y_2 y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3g_1^2 \overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$	$\frac{y e \overline{y_1 y_2 y e}}{4MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 6

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{MF}$
T2	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{11613}}^{(1)}$

$16\pi^2\mu\frac{dC_{11613}}{d\mu}=\beta_{C_{11613}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ie\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{22MF}$	$\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ie\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{11614}}^{(1)}$

$16\pi^2\mu\frac{dC_{11614}}{d\mu}=\beta_{C_{11614}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11\sqrt{3}g_2^2y_1y_2}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1}y_2}{22MF}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3\lambda}12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3\lambda}12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{11711}}^{(1)}$

$$16\pi^2\,\mu\frac{dC_{11711}}{d\mu}=\beta_{C_{11711}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^c}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^c}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{3MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

$\beta_{C_{11712}}^{(1)}$

$16\pi^2\mu\frac{dC_{11712}}{d\mu}=\beta_{C_{11712}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$	$-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1}y_2}{6MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$\beta_{C_{11715}}^{(1)}$

$16\pi^2\mu\frac{dC_{11715}}{d\mu}=\beta_{C_{11715}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ie\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{22MF}$	$\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ie\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{11716}}^{(1)}$

$16\pi^2\mu\frac{dC_{11716}}{d\mu}=\beta_{C_{11716}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11\sqrt{3}g_2^2y_1y_2}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3\lambda}12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3\lambda}12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{11811}}^{(1)}$

$16\pi^2\mu\frac{dC_{11811}}{d\mu}=\beta_{C_{11811}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}\lambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{11}\lambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}\lambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y e}{12 M F}$

$\beta_{C_{11812}}^{(1)}$

$16\pi^2\mu\frac{dC_{11812}}{d\mu}=\beta_{C_{11812}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^c}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^c}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{3MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

$\beta_{C_{11815}}^{(1)}$

$$16\pi^2\mu\frac{dC_{11815}}{d\mu}=\beta_{C_{11815}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{11816}}^{(1)}$

$16\pi^2\mu\frac{dC_{11816}}{d\mu}=\beta_{C_{11816}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda_{12}I\nu v2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\lambda_{12}I\nu v2\overline{y_1y_2}}{22MF}$	$\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{11911}}^{(1)}$

$16\pi^2\mu\frac{dC_{11911}}{d\mu}=\beta_{C_{11911}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3}{3^4}\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3}{3^4}\frac{ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3}{3^4}\frac{ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{11912}}^{(1)}$

$$16\pi^2\mu\frac{dC_{11912}}{d\mu}=\beta_{C_{11912}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{11913}}^{(1)}$

$16\pi^2\mu\frac{dC_{11913}}{d\mu}=\beta_{C_{11913}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda 12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}i\lambda 12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y e}{12 M F}$

$\beta_{C_{11914}}^{(1)}$

$16\pi^2\mu\frac{dC_{11914}}{d\mu}=\beta_{C_{11914}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda a_{12} \text{Inv} 2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\lambda a_{12} \text{Inv} 2\overline{y_1 y_2}}{22MF}$	$\frac{49\sqrt{3}g_2^2\overline{y_1 y_2}}{24MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1 y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$\beta_{C_{111011}}^{(1)}$

$_{16\pi^2}^2\mu\frac{dC_{111011}}{d\mu}=\beta_{C_{111011}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{111012}}^{(1)}$

$16\pi^2\,_{\mu}\frac{dC_{111012}}{d\mu}=\beta_{C_{111012}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3}{3^4}\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3}{3^4}\frac{ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3}{3^4}\frac{ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{111013}}^{(1)}$

$_{16\pi^2}^{\mu}\frac{dC_{111013}}{d\mu}=\beta_{C_{111013}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{49\sqrt{3}g_2^2y_1y_2}{24MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}\lambda a_{12}Inv_1\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$	$\frac{\sqrt{3}\lambda a_{12}Inv_1\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{111014}}^{(1)}$

$$16\pi^2\,_{\mu}\frac{dC_{111014}}{d\mu}=\beta_{C_{111014}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda bda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}i\lambda bda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y e}{12 M F}$

$\beta_{C_{1213}}^{(1)}$

$16\pi^2\mu\frac{dC_{1213}}{d\mu}=\beta_{C_{1213}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1214}}^{(1)}$

$16\pi^2\mu\frac{dC_{1214}}{d\mu}=\beta_{C_{1214}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1223}}^{(1)}$

$16\pi^2\mu\frac{dC_{1223}}{d\mu}=\beta_{C_{1223}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1224}}^{(1)}$

$16\pi^2\mu\frac{dC_{1224}}{d\mu}=\beta_{C_{1224}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1259}}^{(1)}$

$_{16\pi^2}\mu\frac{dC_{1259}}{d\mu}=\beta_{C_{1259}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{g_1^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$	$\frac{3g_1^2 \overline{y_1 y_2}}{16MF}$	$-\frac{y e \overline{y_1 y_2 y e}}{8MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{MF}$
T2	$\frac{lambda12Inv2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{6MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{lambda12Inv2y_1y_2}{6MF}$	$-\frac{\sqrt{3}g_2^2y_1y_2}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1y_2}ye}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{16MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2}ye}{24MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{12510}}^{(1)}$

$16\pi^2\mu\frac{dC_{12510}}{d\mu}=\beta_{C_{12510}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y e}}{MF}$
T3	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{iy e\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$\frac{3ig_1^2\overline{y_1y_2}}{16MF}$	$-\frac{iy e\overline{y_1y_2y\bar{e}}}{8MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}iye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{16MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2y\bar{e}}}{24MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$$\beta_{C_{12511}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{12511}}{d\mu}=\beta_{C_{12511}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$$\beta_{C_{12512}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12512}}{d\mu}=\beta_{C_{12512}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$\beta_{C_{12515}}^{(1)}$

$16\pi^2\mu\frac{dC_{12515}}{d\mu}=\beta_{C_{12515}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{15\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$\beta_{C_{12516}}^{(1)}$

$16\pi^2\mu\frac{dC_{12516}}{d\mu}=\beta_{C_{12516}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{15\sqrt{3}g_2^2y_1y_2}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1}y_2}{22MF}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}lambda12Inv1\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$	$\frac{\sqrt{3}lambda12Inv1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{1269}}^{(1)}$

$16\pi^2\mu\frac{dC_{1269}}{d\mu}=\beta_{C_{1269}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \bar{e}}}{12MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3ig_1^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{iy e \overline{y_1 y_2 y e}}{MF}$
T2	$\frac{ilambda 12 I n v 2 \overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{iy e \overline{y_1 y_2 y e}}{8MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{ilambda 12 I n v 2 \overline{y_1 y_2}}{6MF}$	$\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}iye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{16MF}$	$-\frac{\sqrt{3}iye\overline{y_1}y_2\overline{y}e}{24MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{12610}}^{(1)}$

$16\pi^2\mu\frac{dC_{12610}}{d\mu}=\beta_{C_{12610}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{g_1^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{3g_2^2 \overline{y_1 y_2}}{8 M F}$	$\frac{3g_1^2 \overline{y_1 y_2}}{16 M F}$	$-\frac{y e \overline{y_1 y_2 y e}}{8 M F}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{MF}$
T2	$\frac{lambda12Inv2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{6MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{lambda12Inv2y_1y_2}{6MF}$	$-\frac{\sqrt{3}g_2^2y_1y_2}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1y_2}ye}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{16MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2}ye}{24MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$$\beta_{C_{12611}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12611}}{d\mu}=\beta_{C_{12611}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$\beta_{C_{12612}}^{(1)}$

$16\pi^2\mu\frac{dC_{12612}}{d\mu}=\beta_{C_{12612}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$\beta_{C_{12615}}^{(1)}$

$$16\pi^2\mu\frac{dC_{12615}}{d\mu}=\beta_{C_{12615}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

$\beta_{C_{12616}}^{(1)}$

$$16\pi^2\mu\frac{dC_{12616}}{d\mu}=\beta_{C_{12616}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{15\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$\beta_{C_{12711}}^{(1)}$

$16\pi^2\mu\frac{dC_{12711}}{d\mu}=\beta_{C_{12711}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}\overline{\lambda 12Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3g_1^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{3}\overline{\lambda 12Inv2y_1y_2}}{6MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{ye\overline{y_1}y_2\overline{y_2}}{MF}$
T2	$\frac{lambda12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{ye\overline{y_1}y_2\overline{y_2}}{8MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{lambda12Inv2\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$	$\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{24 M F}$	$-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{12712}}^{(1)}$

$16\pi^2\mu\frac{dC_{12712}}{d\mu}=\beta_{C_{12712}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3ig_1^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{iy e \overline{y_1} y_2 \overline{y} e}{MF}$
T2	$\frac{ilambda 12 I n v 2 \overline{y_1} y_2}{MF}$
T3	$\frac{\sqrt{3} i g_2^2 \overline{y_1} y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{iy e \overline{y_1} y_2 \overline{y} e}{8MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{ilambda 12 I n v 2 \overline{y_1} y_2}{6MF}$	$\frac{\sqrt{3} i g_2^2 \overline{y_1} y_2}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{16MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} \textit{ilambda12Inv1}\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} \textit{ilambda12Inv1}\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$$\beta_{C_{12713}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12713}}{d\mu}=\beta_{C_{12713}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$$\beta_{C_{12714}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12714}}{d\mu}=\beta_{C_{12714}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$\beta_{C_{12811}}^{(1)}$

$16\pi^2\mu\frac{dC_{12811}}{d\mu}=\beta_{C_{12811}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{iy_e\overline{y_1y_2y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$\frac{3ig_1^2\overline{y_1y_2}}{16MF}$	$-\frac{iy_e\overline{y_1y_2y_e}}{8MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{16MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2y_1y_2}{24MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i \lambda \text{Inv1} \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} i \lambda \text{Inv1} \overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{12812}}^{(1)}$

$16\pi^2\mu\frac{dC_{12812}}{d\mu}=\beta_{C_{12812}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}\lambda\overline{Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3g_1^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{3}\lambda\overline{Inv2y_1y_2}}{6MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{ye\overline{y_1y_2y\overline{e}}}{8MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{M F}$
T2	$\frac{3^{\frac{3}{4}} g_2^2 y_1 y_2}{M F}$
T3	$\frac{3^{\frac{3}{4}} y e \bar{y}_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{8 M F}$	$\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24 M F}$	$-\frac{3^{\frac{3}{4}} y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$$\beta_{C_{12813}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12813}}{d\mu}=\beta_{C_{12813}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$$\beta_{C_{12814}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12814}}{d\mu}=\beta_{C_{12814}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$\beta_{C_{12911}}^{(1)}$

$$16\pi^2\mu\frac{dC_{12911}}{d\mu}=\beta_{C_{12911}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{15\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y e}{12 M F}$

$\beta_{C_{12912}}^{(1)}$

$16\pi^2\mu\frac{dC_{12912}}{d\mu}=\beta_{C_{12912}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda a_{12} \text{Inv} 2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\lambda a_{12} \text{Inv} 2\overline{y_1 y_2}}{22MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1 y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$\beta_{C_{121011}}^{(1)}$

$_{16\pi^2}{}_{\mu}\frac{dC_{121011}}{d\mu}=\beta_{C_{121011}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{15\sqrt{3}g_2^2y_1y_2}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}\lambda_{12}In v\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$	$\frac{\sqrt{3}\lambda_{12}In v\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{121012}}^{(1)}$

$$16\pi^2\,\mu\,\frac{dC_{121012}}{d\mu}=\beta_{C_{121012}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda bda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{15\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}i\lambda bda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y e}{12 M F}$

$\beta_{C_{2211}}^{(1)}$

$16\pi^2\mu\frac{dC_{2211}}{d\mu}=\beta_{C_{2211}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{2212}}^{(1)}$

$16\pi^2\mu\frac{dC_{2212}}{d\mu}=\beta_{C_{2212}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}i\lambda T_3\overline{y_1y_2}}{MF}$
T2	$\frac{i\lambda T_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}i\lambda T_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda T_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda T_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{2i\lambda T_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda T_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-\frac{2\sqrt{6}i\lambda T_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{2222}}^{(1)}$

$16\pi^2\mu\frac{dC_{2222}}{d\mu}=\beta_{C_{2222}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{2257}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{2257}}{d\mu}=\beta_{C_{2257}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ye y_1 y_2 ye}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{2258}}^{(1)}$

$16\pi^2\mu\frac{dC_{2258}}{d\mu}=\beta_{C_{2258}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{24MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22511}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22511}}{d\mu}=\beta_{C_{22511}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T2	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{11 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22512}}^{(1)}$

$16\pi^2\mu\frac{dC_{22512}}{d\mu}=\beta_{C_{22512}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y\overline{e}}}{MF}$
T2	$\frac{3^{\frac{3}{4}} i\lambda 12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y\overline{e}}}{12MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\frac{3^{\frac{3}{4}} i\lambda 12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1 y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22513}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22513}}{d\mu}=\beta_{C_{22513}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{49\sqrt{3}g_2^2y_1y_2}{24MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{8MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{22514}}^{(1)}$

$16\pi^2\mu\frac{dC_{22514}}{d\mu}=\beta_{C_{22514}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$	$-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda12Inv2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}\lambda12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{11}\lambda12Inv2\overline{y_1}y_2}{22MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$	$\frac{\sqrt{3}\lambda12Inv1\overline{y_1}y_2}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y e}{12 M F}$

$\beta_{C_{2267}}^{(1)}$

$16\pi^2\mu\frac{dC_{2267}}{d\mu}=\beta_{C_{2267}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{2268}}^{(1)}$

$16\pi^2\mu\frac{dC_{2268}}{d\mu}=\beta_{C_{2268}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye \overline{y_1} y_2 y \overline{e}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1} y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ye y_1 y_2 y \overline{e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22611}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22611}}{d\mu}=\beta_{C_{22611}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22612}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22612}}{d\mu}=\beta_{C_{22612}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T2	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{11 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22613}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22613}}{d\mu}=\beta_{C_{22613}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}i\lambda b d a 12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}i\lambda b d a 12Inv1\overline{y_1}y_2}{6MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$\beta_{C_{22614}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22614}}{d\mu}=\beta_{C_{22614}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{49\sqrt{3}g_2^2y_1y_2}{24MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{8MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{2279}}^{(1)}$

$16\pi^2\mu\frac{dC_{2279}}{d\mu}=\beta_{C_{2279}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$\frac{3g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{MF}$
T2	$\frac{lambda12Inv2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{lambda12Inv2y_1y_2}{3MF}$	$-\frac{\sqrt{3}g_2^2y_1y_2}{4MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1y_2}ye}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2}ye}{12MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22710}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22710}}{d\mu}=\beta_{C_{22710}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{iy_e\overline{y_1y_2y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$\frac{3ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{iy_e\overline{y_1y_2y_e}}{4MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}iye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$	$\frac{\sqrt{3}iye\overline{y_1}y_2\overline{y}e}{12MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22711}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22711}}{d\mu}=\beta_{C_{22711}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{22712}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22712}}{d\mu}=\beta_{C_{22712}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{22MF}$	$\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{22715}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22715}}{d\mu}=\beta_{C_{22715}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y e}{12 M F}$

$\beta_{C_{22716}}^{(1)}$

$16\pi^2\mu\frac{dC_{22716}}{d\mu}=\beta_{C_{22716}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda a_{12} \text{Inv} 2 \overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3} g_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\lambda a_{12} \text{Inv} 2 \overline{y_1 y_2}}{22MF}$	$\frac{49\sqrt{3} g_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

$\beta_{C_{2289}}^{(1)}$

$16\pi^2\mu\frac{dC_{2289}}{d\mu}=\beta_{C_{2289}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{3ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{iy e \overline{y_1} y_2 \overline{y} e}{MF}$
T2	$\frac{ilambda 12 I n v 2 \overline{y_1} y_2}{MF}$
T3	$\frac{\sqrt{3} i g_2^2 \overline{y_1} y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{iy e \overline{y_1} y_2 \overline{y} e}{4MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{ilambda 12 I n v 2 \overline{y_1} y_2}{3MF}$	$\frac{\sqrt{3} i g_2^2 \overline{y_1} y_2}{4MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}iye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1}y_2\overline{y}e}{12MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22810}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22810}}{d\mu}=\beta_{C_{22810}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$\frac{3g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}\lambda 12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\lambda 12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{\sqrt{3}\lambda 12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{\lambda 12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1y_2}ye}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2}ye}{12MF}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22811}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22811}}{d\mu}=\beta_{C_{22811}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{22812}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22812}}{d\mu}=\beta_{C_{22812}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{22815}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22815}}{d\mu}=\beta_{C_{22815}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{49\sqrt{3}g_2^2y_1y_2}{24MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}\lambda a_{12}Inv1\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$	$\frac{\sqrt{3}\lambda a_{12}Inv1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{22816}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22816}}{d\mu}=\beta_{C_{22816}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}i\lambda\overline{12Inv2\overline{y_1y_2}}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y e}{12 M F}$

$\beta_{C_{22911}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22911}}{d\mu}=\beta_{C_{22911}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22912}}^{(1)}$

$16\pi^2\mu\frac{dC_{22912}}{d\mu}=\beta_{C_{22912}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22913}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22913}}{d\mu}=\beta_{C_{22913}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ie\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{22MF}$	$\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ie\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{22914}}^{(1)}$

$16\pi^2\mu\frac{dC_{22914}}{d\mu}=\beta_{C_{22914}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11\sqrt{3}g_2^2y_1y_2}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3\lambda}12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3\lambda}12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{221011}}^{(1)}$

$_{16\pi^2}^2\mu\frac{dC_{221011}}{d\mu}=\beta_{C_{221011}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{221012}}^{(1)}$

$$16\pi^2\,\mu\,\frac{dC_{221012}}{d\mu}=\beta_{C_{221012}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{221013}}^{(1)}$

$_{16\pi^2}^{\mu}\frac{dC_{221013}}{d\mu}=\beta_{C_{221013}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambd a12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambd a12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambd a12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambd a12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^c}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^c}}{12MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$

$\beta_{C_{221014}}^{(1)}$

$_{16\pi^2}^2\mu\frac{dC_{221014}}{d\mu}=\beta_{C_{221014}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\text{ilambda}12Inv2\overline{y_1y_2}}{22MF}$	$\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$